Task	LM	Character-level perturbations				Word-level perturbations			
		PPS=1	PPS=2	PPS=3	PPS=4	PPS=1	PPS=2	PPS=3	PPS=4
TC	BERT	−15.8	−17.2	−18.0	−18.3	−8.8	−10.2	−13.1	−13.8
	RoBERTa	−17.1	−17.9	−18.5	−18.9	−9.4	−11.0	−12.7	−13.3
	XLNet	−16.6	−17.4	−19.2	−19.7	−8.0	−10.3	−12.4	**−12.9**
	ELMo	**−8.8**	**−10.0**	**−11.2**	**−11.8**	**−7.3**	**−9.1**	**−11.5**	−13.2
SA	BERT	−17.4	−18.9	−20.1	−21.3	−11.0	−13.5	−15.1	−16.7
	RoBERTa	−18.4	−19.7	−21.2	−21.9	−11.5	−12.9	**−14.2**	**−14.8**
	XLNet	−17.0	−19.4	−20.6	−21.7	**−10.0**	**−12.4**	−14.3	−15.6
	ELMo	**−11.2**	**−14.4**	**−16.0**	**−17.5**	−11.2	−13.8	−14.9	−16.1
NER	BERT	−12.2	−14.6	−16.3	−16.9	−10.8	−12.7	−14.0	−14.9
	RoBERTa	−12.4	−14.0	−14.8	−16.7	−11.0	−12.5	−13.6	−14.3
	XLNet	−12.2	−13.8	−14.5	−15.0	**−10.2**	**−11.9**	**−13.1**	**−14.0**
	ELMo	**−10.2**	**−12.5**	**−13.1**	**−13.8**	−11.3	−13.0	−15.2	−16.1
SS	BERT	−9.8	−11.3	−12.9	−13.6	−10.0	−12.2	−13.8	−14.5
	RoBERTa	−10.4	−11.8	−13.0	−14.3	−10.4	−12.1	−13.2	−14.1
	XLNet	−10.5	−12.0	−13.2	−14.1	**−9.9**	**−11.4**	**−12.8**	**−13.3**
	ELMo	**−9.2**	**−10.6**	**−11.7**	**−13.2**	−10.2	−12.3	−13.9	−15.4
QA	BERT	−12.2	−14.2	−15.0	−16.4	−8.8	−10.2	−12.5	**−13.0**
	RoBERTa	−13.4	−15.1	−15.8	−16.5	−10.1	−11.0	−12.3	−13.5
	XLNet	−12.4	−13.5	−16.4	−18.9	−7.9	**−9.5**	**−11.2**	−13.1
	ELMo	**−7.8**	**−9.7**	**−11.3**	**−12.0**	**−7.5**	−10.1	−12.0	−13.9

Table 5: Absolute decrease in the performance of the language models, on different NLP tasks, for character-level and word-level perturbations, and with different values of the hyperparameter Perturbation per Sample (PPS). For every task and every value of the hyperparameter PPS, the lowest decrease in the performance is shown in bold face.

is also reported in both tables for every NLP task. We performed five separate fine-tuning runs to test if the performance of the NLP models on the original test set and perturbed samples vary between individual runs. Since there was no statistically significant difference between multiple runs (with respect to a t-test with a significance level of p=0.05), we only report the results of the first fine-tuning and testing run. The language models were neither pretrained nor fine-tuned on perturbed samples. The perturbation methods were only applied to the test sets. As the results show, the language models are sensitive to the perturbations and their performance decreases when the input is slightly noisy. However, RoBERTa still performs better than the other models, and ELMo obtains the lowest scores in general.

The results suggest that some language models can handle specific types of perturbation more effectively than other models. ELMo obtains higher scores than BERT and even performs on par with XLNet and RoBERTa on some character-level perturbations. This can be due to its pure character-based representation that enables the model to use morphological clues, leading to a more robust model against character-level noises.

XLNet is shown to handle perturbations to word ordering more efficiently than the others. This can be an effect of the permutation language modelling that may allow the model to still capture the context and perform more accurately when some context words appear in a different order. The results also suggest those models that were pretrained on larger corpora such as RoBERTa and XLNet are more robust when words are replaced by their synonyms. Furthermore, when the negation perturbation has more impact on the task at hand, e.g. sentiment analysis, the models are less stable and handle the noise less efficiently than on other tasks. Observing the results, we can also point out the LSTM-based model, i.e. ELMo, is more sensitive to the order of words in a sample than the transformer-based models.

Table 5 presents the absolute decrease in the performance of the language models for different PPS values in the range [1, 4]. For every language model, the average of absolute decrease in performance is separately reported on character-level and worl-level perturbations for every NLP task. As can be shown, the models are generally more sensitive to character-level perturbations than word-level ones. Perturbed inputs causes the models to make erronous outcomes on the sentiment analysis task more often than on the other tasks. On the other hand, the question answering task suffers less than the other tasks from noisy inputs.

Figure 1 represents six examples for which perturbations to the input led the RoBERTa model

Example 1) Char-level: Misspelling

Text classification: TREC

Input: What is the lenght of the caostline of the state of Alaska ?
Label: *NUMBER:distance*
Predicted: *LOCATION:state*

Example 2) Char-level: Misspelling

Semantic similarity: STS benchmark

Input: A small bird sitting on a branch in witner.
A small bird perched on an icy brnach.
Similarity: *4.2*
Estimated: *3.5*

Example 3) Char-level: Swapping

Question answering: WikiQA

Question: What deos alkali do to liqiuds?
Answer: *Some authors also define an alkali as a base that dissolves in water.*
Predicted: *A solution of a soluble base has a pH greater than 7.*

Example 4) Word-level: Replace with synonym

Text classification: TREC

Input: What is the diam of a golf_game ball ?
Label: *NUMBER:distance*
Predicted: *DESCRIPTION:definition*

Example 5) Word-level: Repetition

Semantic similarity: STS benchmark

Input: A man plays a trumpet trumpet.
A man is playing playing the trumpet.
Similarity: *5.0*
Estimated: *4.3*

Example 6) Word-level: Deletion

Question answering: WikiQA

Question: what is name of national (anthem) song of Switzerland?
Answer: *The Swiss Psalm is the national anthem of Switzerland.*
Predicted: *Since then, it has been frequently sung at patriotic events.*

Figure 1: Six examples of input perturbations from the three NLP tasks for which the RoBERTa model made wrong decisions, but it made correct decisions on the respective original inputs.

to make wrong decisions, but the model made correct decisions on the respective original inputs. As can be seen, examples 1-3 contain minor character-level noise that causes the model make wrong decisions, however, the perturbed text still seems understandable. In example 4, 'diameter' was replaced by 'diam' and 'golf' was replaced by 'golf_game', but the model failed to handle these changes. In example 5, two repetitive words led the model to estimate a lower similarity score, however, the semantic remained unchanged. Finally, example 6 shows how removing a single word led the model to choose a wrong answer.

5.2 User study

We conducted a user study with 20 participants to investigate how understandable the perturbed texts are to humans. We created a set of perturbations by randomly selecting perturbed samples from the datasets used in the experiments. The samples covered all types of character-level and word-level perturbations.

In the first part of the study, each participant was given 30 perturbed samples from those perturbation methods that are not expected to change the text's meaning with respect to the NLP tasks at hand. These are all the character-level perturbations and three word-level perturbations, i.e. *Repetition*, *Singular/plural verbs*, and *Verb tense*. The participants were also given the original text along with every perturbed sample, and were asked to judge if the perturbed text is understandable and still conveys the same meaning. Every sample contained one, two, or three perturbations.

According to the user evaluations, on average, 94% of the perturbed samples from this set were understandable and still conveyed the same meaning as the original text. These results are well in agreement with our discussion in Section 6.2, i.e. the majority of our proposed perturbations can be automatically produced and used without needing human supervision to ensure understandability and consistency.

In the second part of the study, each participant was given 20 $_{\text{perturbed}}$ samples from those perturbation methods that may change the text's meaning or result in meaningless text. They are the rest of word-level perturbations, i.e. *Deletion*, *Replacement with synonyms*, *Negation*, and *Word order*. The participants were also given the original text along with every perturbed sample, and were asked to judge (with respect to the task at hand) if the perturbed text is still meaningful and consistent with the test set label.

According to the user evaluations, on average, 39% of the perturbed samples from this set were still meaningful and consistent with the label, 12%

of the perturbed samples were meaningful but the label should be changed, and 49% of the perturbed samples were no longer meaningful. These results imply that some perturbations need to be monitored, corrected, or filtered to make sure they are understandable, meaningful, and consistent with the test set label. This helps to fairly estimate the robustness of NLP systems to input perturbations.

6 Related work

Typical performance measures such as accuracy, precision, recall, etc. may not properly reflect how NLP systems behave in real-world use cases. This has motivated many studies to devise novel methods for investigating different capabilities and vulnerabilities of text processing systems. Behavioral testing introduces targeted changes to textual inputs to test linguistic capabilities of systems (Ribeiro et al., 2020). Explanations provide simplified representations of what a complex NLP model has learned (Moradi and Samwald, 2021a, b; Ribeiro et al., 2016). This can help to identify biases and errors in NLP models. Adversarial perturbations have been widely studied to assess the robustness of NLP systems against adversarial samples crafted to fool a model (Alshemali and Kalita, 2020; Ren et al., 2019; Zhang et al., 2020). However, adversarial samples resemble a very specific type of noise. Moreover, most of previous work on adversarial perturbation to NLP models focused on misspelling attacks (Jones et al., 2020; Pruthi et al., 2019; L. Sun et al., 2020). The perturbation methods implemented in this paper represented a wide range of noises that an NLP system may face in real-world situations.

Introducing noise and changing textual inputs were already adopted to assess the ability of models in capturing specific linguistic features such as learning syntax-sensitive dependencies (Linzen et al., 2016), for specific NLP tasks such as machine translation (Belinkov and Bisk, 2018), for detecting biases in language models (Prabhakaran et al., 2019), or to identify susceptible entities in text documents (M. Sun et al., 2018). In this paper, we investigated the robustness on a wide range of tasks, and for various types of character-level and word-level noises in text.

7 Conclusion

In this paper, we introduced and implemented a set of non-adversarial perturbation methods that can be used to evaluate the robustness of NLP systems. We extensively investigated the robustness of high-performance neural language models to noisy input texts. The evaluations on various NLP tasks imply that these models are sensitive to different character-level and word-level perturbations to the input, and the models' performance can decrease when the input contains slight noise. The results suggest that it may be too simplistic to only rely on accuracy scores obtained on benchmark datasets when evaluating the robustness of NLP systems.

The proposed perturbations can be used, along with other methodologies such as CheckList, to test how robust and reliable NLP systems can operate in real-world settings. The experimental results demonstrated that the perturbation methods are effective tools for evaluating NLP systems against noisy data. The user study revealed that only few perturbation methods need to be monitored to make sure they produce meaningful and consistent samples. Most of the perturbation methods can be used automatically to produce noisy test samples. They can be also used as a baseline for evaluating adversarial attacks against non-adversarial perturbations.

Future work may include helping users assess meaning preservation and grammatical correctness in a semi-automatic manner. Sentence encoders such as InferSent (Conneau et al., 2017), Universal Sentence Encoder (Cer et al., 2018), and BERT trained for semantic similarity (Reimers and Gurevych, 2019) can be used to give users clues how semantically similar the original and perturbed sentences are. Moreover, users can be provided with information about grammatical errors in the perturbed text using LanguageTool (Naber, 2003) or other grammar checking tools.

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